Order Derivatives Recover Hidden Polylogarithmic Transseries

Exact completion, resonance jets, and finite reflection rigidity

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Abstract

We study shifted Lambert–polylogarithm sums $F_s(t,a) = \sum_{\ell\geq 0} \mathrm{Li}_s(\mathrm{e}^{-(\ell+a)t})$. At each positive integer order m, a reflection combination has a finite algebraic expansion but a generally nonzero, exactly computable exponential remainder. Its first order derivative restores a divergent series with double Borel poles. With an explicit Borel normalization, the Stokes discontinuity of this derivative is exactly $\pi\mathrm{i}$ times the hidden remainder. We also identify the derivative itself: its linear median requires a convergent dual correction with logarithmic divisor weights. This identification is proved by Mellin inversion, not inferred from the discontinuity.

For arbitrary finite sets of real shifts, we classify all terminating integer-order tails and give finite Fourier–Möbius certificates. A nonzero reflection-compatible measure on r atoms has a hidden action among the first r lattice actions; midpoint grids show the bound is sharp. More generally, the joint kernel of the zeroth and first order tails, on finite measures, is precisely the span of Lebesgue measure. An all-order collision formula shows how reflection removes the explicit Stieltjes term and lowers the logarithmic degree at every order jet. Proofs, a source audit, and reproducible exact and numerical checks are included. Scalar Lambert resurgence and the order-one q-product completion are credited to earlier literature and repository work.

Status. Conventional mathematical proofs are supplied. The results have not been independently refereed or proof-assistant formalized. The intended contribution is the combined all-order, all-real-shift formulation and its finite rigidity consequences; first-priority in the literature is not asserted.

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1 The problem and the contribution

A finite asymptotic expansion need not determine an analytic function. In the family studied here, reflection can annihilate every coefficient of a divergent tail while leaving infinitely many exponentially small terms. The point of this report is to recover those terms from a *specified analytic deformation*: differentiation in the polylogarithm order. We prove both the recovery formula and the analytic completion of the differentiated function.

The variables and the principal observables are

$$F_s(t,a) = \sum_{\ell\geq 0} \mathrm{Li}_s(\mathrm{e}^{-(\ell+a)t}) = \sum_{k\geq 1} \frac{\mathrm{e}^{-akt}}{k^s(1-\mathrm{e}^{-kt})}, \qquad t>0, \quad 0<a<1, \tag{1.1}$$

$$\varepsilon_m = (-1)^{m+1}, \qquad m\in\mathbb{N}, \quad m\geq 1,$$

$$K_m(t,a) = F_m(t,a) + \varepsilon_m F_m(t,1-a), \tag{1.2}$$

$$J_m(t,a) = \partial_s \left(F_s(t,a) + \varepsilon_m F_s(t,1-a)\right)\big|_{s=m}. \tag{1.3}$$

The sign ε_m is held fixed during differentiation. Let P_m denote the explicit finite Laurent polynomial in Theorem 5.1, and let $P_m^{[1]}$ denote the finite power–logarithmic part in Proposition 6.1. The main relations have the form

$$K_m = P_m + N_m, \tag{1.4}$$

$$J_m = P_m^{[1]} + t^m \mathcal{M}\widetilde{J}_m + \Xi_m, \tag{1.5}$$

$$t^m(\mathcal{S}_+ - \mathcal{S}_-)\widetilde{J}_m = \pi \mathrm{i} N_m. \tag{1.6}$$

Here $\widetilde{J}_m$ is a Gevrey-1 formal series with an explicitly meromorphic Borel transform. Both N_m and Ξ_m are convergent series in $Q = \mathrm{e}^{-4\pi^2/t}$; the coefficients of Ξ_m include $\log t$. The linear median $\mathcal{M} = (\mathcal{S}_+ + \mathcal{S}_-)/2$ is defined only for the displayed linear summation problem. No claim about nonlinear median operations is needed.

1.1 What is classical, and what is developed here

The polylogarithm, Hurwitz-zeta special values, Bernoulli Fourier expansions, and the zeta functional equations used below are classical [1]. Dorigoni–Kleinschmidt analyze Lambert-series resurgence, including the survival of nonperturbative terms when an odd-integer asymptotic tail vanishes [2, Section 2.2]. Thus neither the existence of hidden exponential terms nor the general strategy of deforming a parameter is being presented as a new phenomenon.

Fantini–Rella study q-Pochhammer resurgence and character-weighted combinations [3]. A ProveIt report already gives an exact Mellin–Borel completion for the order-one shifted q-factorial and a finite Fourier–Möbius criterion for rational weights [4]. Since $F_1(t,a) = -\log(\mathrm{e}^{-at};\mathrm{e}^{-t})_\infty$, that report directly overlaps the $m=1$ background of this one.

The development here is the explicit first-order-jet completion for all $m\geq 1$ and all real $0<a<1$, the normalization-sensitive identity (1.6), arbitrary-real-atomic reflection rigidity, its sharp action-detection theorem, and the general-measure one-jet kernel. The all-order collision formula supplies a separate bridge to Hurwitz-zeta derivatives, harmonic numbers, and Stieltjes constants. These are proved independently below. A limited literature comparison cannot certify that every formula is absent from all earlier work.

1.2 Repository boundary

The detailed comparison with [4] is pinned to

7bd1ce58abe522b83cd3368bbafa1be97837b75d.

The Polylogarithms index, the Transseries index, and the incoming-directory instructions were also inspected [5]. The binary incoming archives were inventoried but not unpacked in this session. The 552-page combined Polylogarithms manuscript and the whole Transseries collection were not audited. In particular, no claim is made to resolve the separate S6/S8 polylogarithmic conjectures or to review every incoming resonance manuscript. Section 11 records a narrow cross-reference correction and distinguishes it from mathematical corrections required by the new double-pole problem.

2 Analytic setup and Mellin inversion

We use $B_n(a)$ defined by

$$\frac{z\mathrm{e}^{az}}{\mathrm{e}^z-1}=\sum_{n\ge0}B_n(a)\frac{z^n}{n!},\quad B_n=B_n(0),\quad B_1=-\tfrac12. \tag{2.1}$$

Write $H_n^{(r)}=\sum_{k=1}^n k^{-r}$, $H_n=H_n^{(1)}$, and $H_0^{(r)}=0$. The Stieltjes convention is

$$\zeta(1+u)=\frac1u+\sum_{j\ge0}\frac{(-1)^j\gamma_j}{j!}u^j,\qquad \gamma_0=\gamma. \tag{2.2}$$

A prime on $\zeta(w,a)$ always denotes differentiation in its first argument, not in a.

Lemma 2.1 (Analyticity and Mellin representation)**.** *For real $0<a<1$, $F_s(t,a)$ is entire in s and holomorphic in $\Re t>0$. For $t>0$ and $c>\max(1,1-\Re s)$,*

$$F_s(t,a)=\frac{1}{2\pi\mathrm{i}}\int_{(c)}\Gamma(w)\zeta(w,a)\zeta(s+w)t^{-w}\,\mathrm{d}w. \tag{2.3}$$

Every fixed order derivative is obtained by replacing $\zeta(s+w)$ with its corresponding derivative. These operations are uniform on compact subsets of the stated domains, including real a-intervals strictly inside $(0,1)$.

Proof. The second series in (1.1) and each of its s-derivatives are normally convergent on such compact sets: their terms are bounded by $C\mathrm{e}^{-\delta k}k^M(1+\log k)^r$ for suitable $\delta>0$ and M. Expanding $(1-\mathrm{e}^{-kt})^{-1}$ gives the first series and the absolutely convergent double exponential sum. Insert

$$\mathrm{e}^{-x}=\frac{1}{2\pi\mathrm{i}}\int_{(c)}\Gamma(w)x^{-w}\,\mathrm{d}w$$

into that sum. The two Dirichlet series converge absolutely when $c>1$ and $c+\Re s>1$. Exponential vertical decay of Γ justifies Fubini and differentiation. This proves (2.3). □

The standard identities needed later are

$$\zeta(-k,a)=-\frac{B_{k+1}(a)}{k+1},\qquad k\ge0, \tag{2.4}$$

$$B_n(1-a)=(-1)^nB_n(a), \tag{2.5}$$

$$\frac{B_n(a)}{n!}=-\sum_{d\ne0}\frac{\mathrm{e}^{2\pi\mathrm{i}da}}{(2\pi\mathrm{i}d)^n},\qquad n\ge2, \tag{2.6}$$

$$\zeta(w,a)=2\Gamma(1-w)(2\pi)^{w-1}\left(\sin\frac{\pi w}{2}\,C_{1-w}(a)+\cos\frac{\pi w}{2}\,S_{1-w}(a)\right),\quad \Re w<0, \tag{2.7}$$

$$\zeta(z)=2^z\pi^{z-1}\sin\frac{\pi z}{2}\Gamma(1-z)\zeta(1-z), \tag{2.8}$$

where $C_v(a) = \sum_{d\geq 1} \cos(2\pi da)d^{-v}$ and $S_v(a) = \sum_{d\geq 1} \sin(2\pi da)d^{-v}$ whenever these series converge absolutely. The displayed applications of (2.7) are on lines where they do. Equations (2.4)–(2.8) are standard [1, Sections 24.8, 25.4, 25.11].

Contour shifts to any fixed negative nonintegral line are justified by vertical-strip bounds for the zeta functions and their derivatives, together with the exponential factor from $\Gamma(w)$. They give Poincaré expansions as $t \to 0^+$, with a remainder estimated by the new vertical integral. For complex t, the same argument works on closed subsectors of $|\arg t| < \pi/2$. We do not obtain order derivatives by differentiating a nonuniform, noninteger-order expansion: we first differentiate the convergent Mellin integral.

3 Collision jets and Stieltjes cancellation

At $s = m \geq 1$, the pole of $\zeta(s+w)$ meets the Gamma pole at $w = 1-m$. This collision controls the logarithmic part of the expansion. The following coefficient formula handles every derivative order at once.

Theorem 3.1 (All-order collision residue)**.** *Set $n = m-1$, $L = \log t$, and*

$$B_{n,a}(u) = u\Gamma(-n+u)\zeta(-n+u, a). \tag{3.1}$$

This function is holomorphic at $u = 0$. The residue at $w = -n$ in the Mellin representation of $\partial_s^r F_s(t,a)|_{s=m}$ is

$$(-1)^r t^n \left\{ r![u^{r+1}](\mathrm{e}^{-uL}B_{n,a}(u)) + \gamma_r B_{n,a}(0) \right\}, \qquad r \geq 0. \tag{3.2}$$

In particular, this is a polynomial in L of degree at most $r+1$.

Proof. The singular and constant terms of the order derivative are

$$\zeta^{(r)}(1+u) = \frac{(-1)^r r!}{u^{r+1}} + (-1)^r\gamma_r + O(u).$$

The remaining factors are $t^n\mathrm{e}^{-uL}B_{n,a}(u)/u$. The singular term selects the coefficient of u^{r+1}, while the constant term selects $B_{n,a}(0)$. Every $O(u)$ term has zero residue. This proves (3.2), including all signs and factorials. □

For reflection, introduce

$$Z_m(w,a) = \zeta(w,a) + \varepsilon_m\zeta(w,1-a). \tag{3.3}$$

By (2.4) and (2.5), $Z_m(1-m,a) = 0$. Consequently the Gamma singularity in

$$A_{m,a}(u) = \Gamma(1-m+u)Z_m(1-m+u,a) \tag{3.4}$$

is removable.

Corollary 3.2 (Reflection removes one logarithmic degree)**.** *For the fixed-parity order jet $\partial_s^r(F_s(t,a) + \varepsilon_m F_s(t,1-a))|_{s=m}$, the collision contribution is exactly*

$$(-1)^r r! t^{m-1}[u^r](\mathrm{e}^{-uL}A_{m,a}(u)). \tag{3.5}$$

There is no explicit $\gamma_r B(0)$ term, and the logarithmic degree is at most r. Its coefficient of L^r is $t^{m-1}A_{m,a}(0)$.

Proof. For the reflected combination, the function in (3.1) is $uA_{m,a}(u)$, whose constant term is zero. Substitute into Theorem 3.1. The coefficient of L^r comes from $(-uL)^r/r!$; its sign cancels the external $(-1)^r$. □

This cancellation refers to the explicit Stieltjes constants contributed by the pole at 1. It does not say that Hurwitz-zeta derivatives cannot be reexpressed in bases involving other constants. It also does not remove all logarithms: a first jet usually has a $t^{m-1}\log t$ term.

3.1 Harmonic-number generation of the collision coefficients

The finite Gamma factor has the convergent local expansion

$$u\Gamma(-n+u) = \frac{(-1)^n}{n!}\exp\left((H_n-\gamma)u + \sum_{r\geq 2}\frac{H_n^{(r)}+(-1)^r\zeta(r)}{r}u^r\right), \qquad |u|<1. \tag{3.6}$$

Indeed, the left side equals $(-1)^n\Gamma(1+u)\prod_{k=1}^n(1-u/k)^{-1}/n!$. Taking its local logarithm proves (3.6). Together with the Taylor coefficients of $\zeta(-n+u,a)$, this gives a finite, algorithmic expression for every collision jet. Only finitely many harmonic sums and Hurwitz-zeta derivatives are required at each order.

For example, with $Z_j = Z_m^{(j)}(1-m,a)$ and $h = H_{m-1}-\gamma$,

$$\begin{aligned}
A_{m,a}(0) &= \frac{(-1)^{m-1}}{(m-1)!}Z_1,\\
A'_{m,a}(0) &= \frac{(-1)^{m-1}}{(m-1)!}\left(\frac{Z_2}{2}+hZ_1\right),\\
[u^2]A_{m,a}(u) &= \frac{(-1)^{m-1}}{(m-1)!}\left(\frac{Z_3}{6}+\frac{hZ_2}{2}+\frac{h^2+H_{m-1}^{(2)}+\zeta(2)}{2}Z_1\right).
\end{aligned} \tag{3.7}$$

The second-order reflected collision is therefore $t^{m-1}(A(0)L^2-2A'(0)L+A''(0))$. This exact formula already reaches the Gamma/polygamma and Stieltjes part of the Polylogarithms program without conjectural constant reduction.

4 The ordinary integer-order Borel kernel

Before imposing reflection, the integer-order expansion of a single shift is

$$F_m(t,a) \sim P_m^F(t,a) + t^{m-1}\widetilde{R}_{m,a}(t), \qquad \widetilde{R}_{m,a}(t) = \sum_{j\geq 1} c_{m,j}(a)t^{2j}, \tag{4.1}$$

where

$$\begin{aligned}
P_m^F(t,a) = {}& \frac{\zeta(m+1)}{t} + \sum_{\substack{0\leq k\leq m\\ k\neq m-1}}\frac{(-t)^k}{k!}\zeta(-k,a)\zeta(m-k)\\
&+ \frac{(-t)^{m-1}}{(m-1)!}\left(\zeta'(1-m,a)+(H_{m-1}-\log t)\zeta(1-m,a)\right),
\end{aligned} \tag{4.2}$$

$$c_{m,j}(a) = \frac{(-1)^{m-1}B_{m+2j}(a)B_{2j}}{(m+2j)!\,2j}. \tag{4.3}$$

These formulas follow by shifting (2.3). The pole at $w=1$ gives the t^{-1} term, the collision is Theorem 3.1 with $r=0$, and the trivial zeta zeros leave only the displayed tail.

4.1 Normalization and continuation

For a formal power series with zero constant term we use

$$\mathcal{B}(t^r) = \frac{\xi^{r-1}}{(r-1)!}, \qquad \mathcal{L}_\theta b(t) = \int_0^{\mathrm{e}^{\mathrm{i}\theta}\infty} \mathrm{e}^{-\xi/t} b(\xi)\,\mathrm{d}\xi. \tag{4.4}$$

There is no factor $1/t$ in the Laplace transform. Let

$$A_N = 4\pi^2 N, \tag{4.5}$$

$$D_m(N,a) = \sum_{d|N} \frac{\mathrm{e}^{2\pi\mathrm{i}da} + (-1)^m \mathrm{e}^{-2\pi\mathrm{i}da}}{d^m}. \tag{4.6}$$

Theorem 4.1 (Meromorphic simple-pole kernel)**.** *The Borel transform of $\widetilde{R}_{m,a}$ is*

$$\mathcal{B}\widetilde{R}_{m,a}(\xi) = \frac{2(-1)^{m-1}}{(2\pi\mathrm{i})^m} \sum_{N\geq 1} D_m(N,a) \frac{\xi}{A_N^2 - \xi^2}. \tag{4.7}$$

The series is normally convergent off the lattice $\{\pm A_N : N \geq 1\}$. At either $\xi = A_N$ or $\xi = -A_N$, its residue is

$$\frac{(-1)^m}{(2\pi\mathrm{i})^m} D_m(N,a). \tag{4.8}$$

A displayed pole is removable exactly when this coefficient vanishes. The formal tail is Gevrey-1 and is Borel–Laplace summable in every nonreal direction that is compatible with the Laplace exponential.

Proof. Insert (2.6) and $B_{2j} = (-1)^{j+1} 2(2j)!\zeta(2j)/(2\pi)^{2j}$ into (4.3). Expand $\zeta(2j)$ as a Dirichlet series, pair the positive and negative Fourier indices, and sum

$$\sum_{j\geq 1} \frac{\xi^{2j-1}}{(4\pi^2 dn)^{2j}} = \frac{\xi}{(4\pi^2 dn)^2 - \xi^2}.$$

This gives (4.7) near zero, with $N = dn$. For every N, $|D_m(N,a)| \leq 2\sum_{d|N} d^{-m} \leq 2(1+\log N)$. On a compact set, the large-N summands are $O((1+\log N)N^{-2})$, proving normal convergence and meromorphic continuation. Each rational summand has residue $-1/2$ at both of its poles, which gives (4.8).

On a ray separated from the real axis, $|A_N^2 - \xi^2| \geq c_\theta(A_N^2 + |\xi|^2)$. Split the sum at $N \asymp 1 + |\xi|$ to obtain at most logarithmic growth at infinity, and $O(\xi)$ at zero. Thus the corresponding Laplace integrals converge, and their termwise evaluation is justified. Analyticity of the Borel germ in $|\xi| < 4\pi^2$ gives the Gevrey-1 coefficient estimate by Cauchy's inequality. Expanding the integrand near zero and estimating the remainder of the Laplace integral gives the stated formal expansion of its sum. □

Reflection removes this entire kernel, since its Bernoulli coefficient has degree $m + 2j$ and $B_{m+2j}(a) + \varepsilon_m B_{m+2j}(1-a) = 0$. It is essential not to conclude from this that the reflected analytic function equals its finite polynomial.

5 Exact reflection and a convergent dual polylogarithm

Define

$$\begin{aligned} h_m(d,a) &= \mathrm{e}^{2\pi \mathrm{i} d a} + \varepsilon_m \mathrm{e}^{-2\pi \mathrm{i} d a}, \\ E_m(N,a) &= \sum_{d \mid N} \frac{h_m(d,a)}{d^m}, \qquad (5.1) \\ \Phi_m(Q,a) &= \sum_{N \geq 1} E_m(N,a) Q^N \\ &= \sum_{\ell \geq 1} \left(\mathrm{Li}_m(\mathrm{e}^{2\pi \mathrm{i} a} Q^\ell) + \varepsilon_m \, \mathrm{Li}_m(\mathrm{e}^{-2\pi \mathrm{i} a} Q^\ell) \right), \quad |Q| < 1. \qquad (5.2) \end{aligned}$$

These are normally convergent power series. The factor $(-1)^m$ in D_m has changed to $\varepsilon_m = (-1)^{m+1}$ in E_m: the two divisor sequences encode complementary reflection components.

Theorem 5.1 (Exact reflected duality)**.** *For every integer $m \geq 1$, real $0 < a < 1$, and $t > 0$,*

$$K_m(t,a) = P_m(t,a) + \left(\frac{\mathrm{i}t}{2\pi} \right)^{m-1} \Phi_m(\mathrm{e}^{-4\pi^2/t}, a), \qquad (5.3)$$

where the finite Laurent polynomial is

$$\begin{aligned} P_m(t,a) = \frac{(1+\varepsilon_m)\zeta(m+1)}{t} &+ \sum_{\substack{0 \leq k \leq m \\ k \neq m-1}} \frac{(-t)^k}{k!} \zeta(m-k) Z_m(-k,a) \\ &+ \frac{(-t)^{m-1}}{(m-1)!} Z_m'(1-m,a). \qquad (5.4) \end{aligned}$$

In particular, $N_m = K_m - P_m$ is given exactly by the convergent dual series, not merely by its first exponential term.

Proof. In the Mellin integral for K_m, the zeta factor $\zeta(w,a)$ is replaced by $Z_m(w,a)$. Shift to $\Re w = -m - \delta$, where $0 < \delta < 1$. The crossed residues are precisely (5.4): the collision loses its logarithm because $Z_m(1-m,a) = 0$.

Put $v = 1 - m - w$ and

$$H_m(v,a) = \sum_{d \geq 1} \frac{h_m(d,a)}{d^{m+v}}.$$

Combining (2.7), (2.8), and $\Gamma(w)\Gamma(1-w) = \pi / \sin(\pi w)$ gives

$$\Gamma(w)\zeta(m+w) Z_m(w,a) = \left(\frac{\mathrm{i}}{2\pi} \right)^{m-1} (4\pi^2)^{-v} \Gamma(v)\zeta(v) H_m(v,a). \qquad (5.5)$$

For clarity, when m is odd, Z_m keeps the cosine Fourier series; when m is even, it keeps the sine Fourier series, with $H_m = 2\mathrm{i} S_{m+v}$. In either case the elementary trigonometric factors in (5.5) reduce to the displayed constant. The identity holds across any removable zeros by analytic continuation.

The reflected remainder is consequently

$$\left(\frac{\mathrm{i}t}{2\pi} \right)^{m-1} \frac{1}{2\pi \mathrm{i}} \int_{(1+\delta)} \Gamma(v)\zeta(v) H_m(v,a) X^{-v} \, \mathrm{d}v, \qquad X = \frac{4\pi^2}{t}. \qquad (5.6)$$

Both Dirichlet series converge absolutely on this line. Termwise Mellin inversion evaluates (5.6) as

$$\left(\frac{\mathrm{i}t}{2\pi}\right)^{m-1}\sum_{d,n\geq 1}\frac{h_m(d,a)}{d^m}\mathrm{e}^{-Xdn},$$

which is (5.3). All sums in this last expression converge absolutely for every $t>0$. □

Remark 5.2 (An exact transseries, not only an asymptotic statement). The action set is $4\pi^2\mathbb{N}$. It is locally finite, and the exponential block is an ordinary convergent power series in Q. Its coefficients are explicit divisor sums. The identity therefore supplies an analytic realization of the full exponentially small block, while the ordinary power expansion contains none of that block's coefficients.

6 The order derivative and its double-pole kernel

Proposition 6.1 (Finite part and activated coefficients). *The fixed-parity first jet* (1.3) *has the expansion*

$$J_m(t,a)\sim P_m^{[1]}(t,a)+t^m\widetilde{J}_m(t,a),\qquad \widetilde{J}_m(t,a)=\sum_{j\geq 1}d_{m,j}(a)t^{2j},\tag{6.1}$$

where

$$\begin{aligned}P_m^{[1]}(t,a)&=\frac{(1+\varepsilon_m)\zeta'(m+1)}{t}+\sum_{\substack{0\leq k\leq m\\ k\neq m-1}}\frac{(-t)^k}{k!}\zeta'(m-k)Z_m(-k,a)\\&\quad+\frac{(-t)^{m-1}}{(m-1)!}\left((\log t-H_{m-1}+\gamma)Z_m'(1-m,a)-\frac{1}{2}Z_m''(1-m,a)\right),\end{aligned}\tag{6.2}$$

$$d_{m,j}(a)=\frac{(-1)^{m+j+1}(2j)!\,B_{m+2j+1}(a)\zeta(2j+1)}{(m+2j+1)!(2\pi)^{2j}}.\tag{6.3}$$

Proof. Differentiate the Mellin integral first and then shift its contour. The finite residues away from $w=1-m$ give the first line of (6.2). Corollary 3.2 with $r=1$ gives the second line.

For $k>m$, the coefficient at t^k is $(-1)^kZ_m(-k,a)\zeta'(m-k)/k!$. When $k=m+2j-1$, the Bernoulli reflection makes $Z_m(-k,a)=0$. When $k=m+2j$, the zero of $\zeta(-2j)$ is replaced by its nonzero derivative. Differentiating (2.8) at $-2j$ gives

$$\zeta'(-2j)=\frac{(-1)^j(2j)!}{2(2\pi)^{2j}}\zeta(2j+1),\qquad j\geq 1.\tag{6.4}$$

Also $Z_m(-m-2j,a)=-2B_{m+2j+1}(a)/(m+2j+1)$. Substitution proves (6.3). □

The entire activated tail is therefore driven by derivatives at the trivial zeros of the Riemann zeta function. This explains both its odd zeta values and its one-step change of power parity.

Theorem 6.2 (Explicit double-pole Borel transform). *Let* $C_m=(-1)^m/(2\pi\mathrm{i})^{m+1}$. *Then*

$$\mathcal{B}\widetilde{J}_m(\xi,a)=C_m\sum_{N\geq 1}\frac{E_m(N,a)}{N}\frac{2A_N^2\xi}{(A_N^2-\xi^2)^2}.\tag{6.5}$$

This is normally convergent off the action lattice and is of at most polynomial growth on rays separated from the real axis. At $\xi=A_N$, *the coefficient of* $(\xi-A_N)^{-2}$ *is*

$$2\pi^2C_mE_m(N,a),\tag{6.6}$$

and the coefficient of $(\xi-A_N)^{-1}$ *is zero. The double-pole coefficient at* $-A_N$ *is the negative of* (6.6). *The series* $\widetilde{J}_m$ *is Gevrey-1 and summable off the real Borel rays.*

Proof. Apply the Bernoulli Fourier expansion to (6.3) and expand $\zeta(2j+1)$. After division by $(2j-1)!$, the extra factor is $2j$. For $A \neq 0$,

$$\sum_{j\geq 1} 2j \frac{\xi^{2j-1}}{A^{2j}} = \frac{2A^2\xi}{(A^2-\xi^2)^2}. \tag{6.7}$$

The leftover divisor weight is d^{-m}/N, giving (6.5) with the stated constant. The large-N bound on compact sets is now $O((1+\log N)N^{-3})$. On an allowed ray, use $|A_N^2 - \xi^2| \geq c(A_N^2 + |\xi|^2)$ and split at $N \asymp 1 + |\xi|$; this proves polynomial growth and Laplace convergence.

Finally,

$$\frac{2A^2\xi}{(A^2-\xi^2)^2} = \frac{\mathrm{d}}{\mathrm{d}\xi}\frac{\xi^2}{A^2-\xi^2}. \tag{6.8}$$

A derivative of a meromorphic function has zero residue. Its double coefficients at A and $-A$ are $A/2$ and $-A/2$, respectively. Multiplication by $C_m E_m(N,a)/N$ proves (6.6). The same Cauchy and Laplace arguments as in Theorem 4.1 give the Gevrey and summability assertions. □

A zero residue of the *Borel kernel* does not mean a zero Stokes jump. The residue must be taken after multiplication by the Laplace exponential; that operation turns each double pole into a nonzero simple-pole coefficient. The next section computes it with its sign.

7 Exact order-jet completion and recovery of the hidden block

For $t > 0$, define $\mathcal{S}_+$ and $\mathcal{S}_-$ by Laplace rays immediately above and below the positive Borel axis, respectively. More precisely, one can use fixed rays of angles θ and $-\theta$, where $0 < \theta < \pi/2$, and deform within the indicated half-planes. Their values are independent of the particular such angle. The difference of the upper and lower contours is clockwise around positive poles.

7.1 The finite-part kernel needed for a double pole

Lemma 7.1 (A simple-pole representation of the linear median)**.** *For $\lambda > 0$, let*

$$I(\lambda) = \mathrm{PV}\int_0^\infty \mathrm{e}^{-\lambda y}\frac{y^2}{1-y^2}\,\mathrm{d}y = -\frac{1}{\lambda} + \frac{1}{2}\left(\mathrm{e}^{-\lambda}\,\mathrm{Ei}(\lambda) + \mathrm{e}^{\lambda}E_1(\lambda)\right). \tag{7.1}$$

It is $O(\lambda^{-1}+1)$ at zero and $O(\lambda^{-3})$ at infinity. For $1 < \Re v < 3$,

$$\int_0^\infty \lambda^{v-1} I(\lambda)\,\mathrm{d}\lambda = \frac{\pi}{2}\Gamma(v)\tan\frac{\pi v}{2}. \tag{7.2}$$

For $A > 0$, $t > 0$, and $\lambda = A/t$, the upper and lower Laplace integrals of the double-pole kernel satisfy

$$\int_0^{\mathrm{e}^{\pm \mathrm{i}\theta}\infty} \mathrm{e}^{-\xi/t}\frac{2A^2\xi}{(A^2-\xi^2)^2}\,\mathrm{d}\xi = \lambda I(\lambda) \;\pm\; \frac{\pi \mathrm{i}\lambda}{2}\mathrm{e}^{-\lambda}. \tag{7.3}$$

Proof. Use $y^2/(1-y^2) = -1 + 1/(1-y^2)$ and partial fractions. The elementary exponential-integral evaluations give (7.1). They give the endpoint estimates as well: in particular $I(\lambda) \sim \sum_{j\geq 1}(2j)!\lambda^{-2j-1}$ at infinity. This last expansion is asymptotic, not a convergent series.

For the Mellin transform, first omit $(1-\delta, 1+\delta)$ from the y-integral. Away from the pole, the absolute integral after integrating in λ converges at both endpoints precisely when $1 < \Re v < 3$. Subtracting the value of the pole numerator at $y = 1$ shows that the difference between the truncated and principal-value kernels is bounded by $C\delta(1+\lambda)\mathrm{e}^{-\lambda/2}$ for $0 < \delta < 1/2$. This

bound is integrable against $\lambda^{\Re v-1}\,\mathrm{d}\lambda$. Consequently the limit commutes with the Mellin integral, giving

$$\begin{aligned}\Gamma(v)\,\mathrm{PV}\int_0^\infty \frac{y^{2-v}}{1-y^2}\,\mathrm{d}y &= \frac{\Gamma(v)}{2}\,\mathrm{PV}\int_0^\infty \frac{u^{(1-v)/2}}{1-u}\,\mathrm{d}u\\ &= \frac{\pi}{2}\Gamma(v)\cot\frac{\pi(3-v)}{2} = \frac{\pi}{2}\Gamma(v)\tan\frac{\pi v}{2}.\end{aligned}$$

The principal-value cotangent identity follows by splitting at 1, inverting the integration variable in the second part, and using the partial fraction expansion of the cotangent. Mellin inversion is valid on any vertical line $1 < c < 3$; the Gamma factor supplies exponential decay there.

For (7.3), use (6.8) and integrate by parts on a nonreal ray. The endpoint contributions vanish. After $\xi = Ay$, the resulting integral is λ times the lateral integral defining I. The residue of $y^2/(1-y^2)$ at $y=1$ is $-1/2$. The upper contour contributes $+\pi\mathrm{i}\mathrm{e}^{-\lambda}/2$ relative to the symmetric principal value, and the lower contour contributes its negative. This proves the formula and fixes its orientation. □

The double-pole Laplace integral itself does *not* possess an ordinary Cauchy principal value on the real line. Its linear median is a Hadamard finite part, equivalently the simple-pole expression in (7.3). This distinction is necessary even though the double-pole kernel has zero residue before the exponential is inserted.

7.2 The exact logarithmic dual correction

Define

$$\Xi_m(t,a) = \left(\frac{\mathrm{i}t}{2\pi}\right)^{m-1}\sum_{d\ge1}\frac{h_m(d,a)}{d^m}\frac{\log(t/(2\pi d))}{\mathrm{e}^{4\pi^2 d/t}-1}. \tag{7.4}$$

For every $t>0$ this series converges absolutely. It is an exact polylogarithmic order-derivative block: if

$$\Phi_{s|m}(Q,a) = \sum_{\ell\ge1}\left(\mathrm{Li}_s(\mathrm{e}^{2\pi\mathrm{i}a}Q^\ell) + \varepsilon_m\,\mathrm{Li}_s(\mathrm{e}^{-2\pi\mathrm{i}a}Q^\ell)\right), \tag{7.5}$$

then

$$\Xi_m(t,a) = \left(\frac{\mathrm{i}t}{2\pi}\right)^{m-1}\left(\log\frac{t}{2\pi}\,\Phi_m(Q,a) + \partial_s\Phi_{s|m}(Q,a)\Big|_{s=m}\right). \tag{7.6}$$

The parity sign, again, is fixed.

Theorem 7.2 (Exact first-jet completion and activation)**.** *Under the hypotheses of Theorem 5.1,*

$$J_m(t,a) = P_m^{[1]}(t,a) + t^m\mathcal{M}\widetilde{J}_m(t,a) + \Xi_m(t,a), \tag{7.7}$$

$$t^m(\mathcal{S}_+ - \mathcal{S}_-)\widetilde{J}_m(t,a) = \pi\mathrm{i}\big(K_m(t,a) - P_m(t,a)\big). \tag{7.8}$$

$$J_m(t,a) = P_m^{[1]}(t,a) + t^m\mathcal{S}_\pm\widetilde{J}_m(t,a) + \Xi_m(t,a) \mp \frac{\pi\mathrm{i}}{2}N_m(t,a). \tag{7.9}$$

The linear median also has the absolutely convergent kernel formula

$$t^m\mathcal{M}\widetilde{J}_m(t,a) = \left(\frac{\mathrm{i}t}{2\pi}\right)^{m-1}\sum_{N\ge1}E_m(N,a)I(A_N/t). \tag{7.10}$$

Proof. Identification of the analytic function. Differentiate (2.3) in s before specializing to $s=m$, and then form the reflected combination. Shift to $\Re w = -m-\delta$ with $0<\delta<1$. Proposition 6.1 computes the crossed residues as $P_m^{[1]}$.

In the remainder set $v = 1 - m - w$ as in Theorem 5.1. Logarithmic differentiation of the Riemann functional equation, wherever none of its factors is zero, gives the multiplier

$$\log(2\pi) - \psi(v) - \frac{\zeta'(v)}{\zeta(v)} + \frac{\pi}{2}\tan\frac{\pi v}{2}.$$

To avoid any artificial division by a zeta zero, put $G(v) = \Gamma(v)\zeta(v)$ and use the holomorphic product expression

$$\log(2\pi)G(v) - G'(v) + \frac{\pi}{2}\tan\frac{\pi v}{2}G(v). \tag{7.11}$$

The identity extends across removable zeros. The exact remainder is

$$\left(\frac{\mathrm{i}t}{2\pi}\right)^{m-1}\frac{1}{2\pi\mathrm{i}}\int_{(1+\delta)}\left(\log(2\pi)G - G' + \frac{\pi}{2}\tan\frac{\pi v}{2}G\right)H_m(v,a)X^{-v}\,\mathrm{d}v. \tag{7.12}$$

There are no tangent poles on this line.

Integrate the $-G'$ term by parts along the vertical contour. The horizontal endpoint terms vanish by Gamma decay. The first two terms of the integrand become

$$G(v)\left(\log\frac{t}{2\pi}H_m(v,a) + H'_m(v,a)\right)X^{-v},$$

since $\log(2\pi) - \log X = \log(t/(2\pi))$. The series for H_m and H'_m converge absolutely. Mellin inversion of their Dirichlet expansions therefore gives exactly (7.4).

For the last term in (7.12), expand $\zeta(v)H_m(v,a) = \sum_{N\geq 1}E_m(N,a)N^{-v}$. Lemma 7.1 identifies the resulting inverse Mellin integral with $I(XN)$. Thus the remainder not accounted for by Ξ_m is the right side of (7.10). This series converges absolutely because $E_m(N,a) = O(1+\log N)$ and $I(XN) = O(N^{-3})$ for fixed $X > 0$.

Identification with the Borel median. Integrate (6.5) termwise on upper and lower rays, which is justified by the absolute majorants in Theorem 6.2. Lemma 7.1 gives their average. The constants simplify using

$$4\pi^2 C_m = \left(\frac{\mathrm{i}}{2\pi}\right)^{m-1}. \tag{7.13}$$

Indeed, $A_N/(tN) = 4\pi^2/t$ in every summand. This proves (7.10), and hence (7.7).

The discontinuity. By (7.3), the difference of the two lateral integrals of one rational summand is $\pi\mathrm{i}(A_N/t)\mathrm{e}^{-A_N/t}$. Multiplying by $t^m C_m E_m(N,a)/N$ and summing gives

$$4\pi^3\mathrm{i}C_m t^{m-1}\sum_{N\geq 1}E_m(N,a)Q^N = \pi\mathrm{i}\left(\frac{\mathrm{i}t}{2\pi}\right)^{m-1}\Phi_m(Q,a).$$

Theorem 5.1 identifies the last factor with N_m. This proves (7.8). Combining it with the median identity gives (7.9). □

Remark 7.3 (A direct residue check). At $\xi = A_N$, write the Borel principal part as $c_N(\xi - A_N)^{-2}$, with $c_N = 2\pi^2 C_m E_m(N,a)$. The residue of the *Laplace integrand* is $-c_N t^{-1}\mathrm{e}^{-A_N/t}$. The upper-minus-lower contour multiplies this by $-2\pi\mathrm{i}$. This reproduces the positive factor in (7.8) and independently explains why simply taking residues of the unweighted Borel kernel would miss the jump.

7.3 Complex sectors and transseries interpretation

All exact identities proved for $t > 0$ continue to $\Re t > 0$ with the principal $\operatorname{Log} t$ and the corresponding analytic continuations of the lateral sums. To see this without manipulating formal symbols, $I(A_N/t)$ is holomorphic there by (7.1); its large-N estimates are uniform on compact subsets. The sums in (7.10), (5.2), and (7.4) are consequently normally convergent. Both sides of the identities are holomorphic and agree on the positive real axis. Alternatively, fixed nonreal Laplace rays give overlapping sectorial definitions of the same continuations.

On a closed subsector $|\arg t| \le \pi/2 - \eta$, $\Re(4\pi^2/t) \ge 4\pi^2 \sin\eta/|t|$. The factors Q^N are then uniformly exponentially small. The exact structure of the first jet is one Borel-summed power block plus convergent sectors $t^{m-1}(\alpha_N \operatorname{Log} t + \beta_N)Q^N$. No accumulating action set or formal-support completion is needed for this particular transseries. The boundary directions $\arg t = \pm\pi/2$ are not covered by this small-exponential assertion.

8 Finite reflection rigidity and action detection

Let

$$\mu = \sum_{j=1}^{r} c_j \delta_{a_j}, \qquad 0 < a_j < 1, \tag{8.1}$$

where the a_j are distinct and the displayed c_j are nonzero. Let $R(a) = 1 - a$, write $R_*\mu$ for the reflected measure, and define

$$F_{s,\mu}(t) = \int F_s(t,a)\,\mathrm{d}\mu(a), \qquad \widehat{\mu}(n) = \int \mathrm{e}^{2\pi \mathrm{i} n a}\,\mathrm{d}\mu(a). \tag{8.2}$$

The atoms need not be rational. By a *terminating tail* we mean that the integer-order power–logarithmic expansion has only finitely many nonzero coefficients. The fixed finite logarithmic collision part is not included in the word "tail."

Theorem 8.1 (Atomic reflection rigidity with a finite certificate)**.** *For $m \ge 1$, the following conditions are equivalent:*

(i) the asymptotic expansion of $F_{m,\mu}$ terminates;

(ii) all its tail coefficients t^{m+2j-1}, $j \ge 1$, vanish;

(iii) $R_*\mu = \varepsilon_m \mu$;

(iv) all positive-pole coefficients in its normalized Borel tail vanish.

Let M be the number of distinct points in $\operatorname{supp}(\mu) \cup R(\operatorname{supp}(\mu))$. For condition (iv), it suffices to check the first M action locations; in particular, at most $2r$ checks suffice.

Proof. The weighted Borel kernel in Theorem 4.1 has divisor coefficients

$$D_{m,\mu}(N) = \sum_{d \mid N} \frac{h_\mu(d)}{d^m}, \quad h_\mu(d) = \widehat{\mu}(d) + (-1)^m \widehat{\mu}(-d). \tag{8.3}$$

Möbius inversion is the exact identity

$$h_\mu(n) = n^m \sum_{d \mid n} \mu_{\mathrm{Mob}}(n/d) D_{m,\mu}(d). \tag{8.4}$$

Thus the vanishing of $D_{m,\mu}(1), \dots, D_{m,\mu}(M)$ implies $h_\mu(1) = \cdots = h_\mu(M) = 0$.

The latter are the first M Fourier coefficients of $\nu = \mu + (-1)^m R_*\mu$. Write this measure on its at most M distinct nodes, with $z_j = \mathrm{e}^{2\pi \mathrm{i} a_j}$. The matrix $(z_j^n)_{1\le n,j\le M}$ is invertible: its determinant is a nonzero product of the z_j and the Vandermonde factors $z_k - z_j$. Padding with zero coefficients if necessary gives $\nu = 0$, which is condition (iii).

Conversely, (iii) and Bernoulli reflection give (ii), and (ii) gives (i). If (i) holds, the Borel germ is a polynomial. Its meromorphic continuation (4.7) must equal that polynomial wherever both are defined; hence every residue vanishes and (iv) holds. This proves all implications and the finite certificate. □

The finite certificate refers to action-pole data, not to an arbitrary list of the first M Taylor coefficients. Those are different finite linear systems. At rational shifts, finite Fourier transforms give a natural alternative presentation; for arbitrary real shifts the Vandermonde proof avoids any rationality hypothesis.

Theorem 8.2 (Hidden actions are visible in one order derivative)**.** *Suppose the nonzero measure* (8.1) *satisfies $R_*\mu = \varepsilon_m \mu$. Then its exact hidden remainder is*

$$N_{m,\mu}(t) = F_{m,\mu}(t) - P_{m,\mu}(t) = \left(\frac{\mathrm{i}t}{2\pi}\right)^{m-1} \sum_{N\ge 1} b_{m,\mu}(N) Q^N, \quad b_{m,\mu}(N) = \sum_{d\mid N} \frac{\widehat{\mu}(d)}{d^m}. \tag{8.5}$$

Here $P_{m,\mu} = \frac{1}{2}\int P_m(t,a)\,\mathrm{d}\mu(a)$. The first nonzero coefficient in (8.5) *occurs at an index $N_0 \le r$. The first order derivative has its first nonzero positive double Borel pole at exactly A_{N_0}, and its jump is $\pi \mathrm{i} N_{m,\mu}$.*

Proof. Since $R_*\mu = \varepsilon_m\mu$, integration of a reflected pair gives $\int K_m(t,a)\,\mathrm{d}\mu(a) = 2F_{m,\mu}(t)$. Also $\widehat{\mu}(-d) = \varepsilon_m \widehat{\mu}(d)$, so that $\frac{1}{2}\int E_m(N,a)\,\mathrm{d}\mu(a) = b_{m,\mu}(N)$. Integrating Theorem 5.1 proves (8.5). The same reasoning applies to the fixed-parity order derivative; Theorems 6.2 and 7.2 identify its double-pole coefficient and its discontinuity.

If $\widehat{\mu}(1),\dots,\widehat{\mu}(r)$ all vanished, the $r\times r$ Vandermonde system for the original atomic measure would force $c_1 = \cdots = c_r = 0$. Choose the first nonzero Fourier coefficient, with index $n_0 \le r$. Every proper-divisor term in $b_{m,\mu}(n_0)$ vanishes, leaving $b_{m,\mu}(n_0) = \widehat{\mu}(n_0)/n_0^m \neq 0$. Earlier b's are zero. This proves $N_0 = n_0 \le r$ and the matching double-pole assertion. □

On prescribed nodes, the hidden coefficients recover the weights by an explicit two-stage linear procedure. First use

$$\widehat{\mu}(n) = n^m \sum_{d\mid n} \mu_{\mathrm{Mob}}(n/d) b_{m,\mu}(d). \tag{8.6}$$

Then solve the Vandermonde system on the known $\mathrm{e}^{2\pi \mathrm{i} a_j}$. This is an exact identification statement; it does not assert numerical stability for nearly colliding nodes.

8.1 Sharpness by midpoint grids

For $r \ge 1$, set

$$\mu_r = \sum_{j=0}^{r-1} \delta_{(j+1/2)/r}. \tag{8.7}$$

This measure is reflection-symmetric. Its Fourier coefficients are

$$\widehat{\mu}_r(n) = \begin{cases} r(-1)^k, & n = rk, \\ 0, & r \nmid n. \end{cases} \tag{8.8}$$

For every odd m, Theorem 8.2 therefore has $N_0 = r$. In particular,

$$N_{m,\mu_r}(t) = -r^{1-m}\left(\frac{\mathrm{i}t}{2\pi}\right)^{m-1} \mathrm{e}^{-4\pi^2 r/t} + O\left(t^{m-1}\mathrm{e}^{-8\pi^2 r/t}\right) \quad (t \to 0^+). \tag{8.9}$$

The first $r-1$ activated poles vanish, so the bound r cannot be improved uniformly. This arithmetic cancellation also has the exact lattice explanation

$$\sum_{j=0}^{r-1} F_s\left(t, \frac{j+1/2}{r}\right) = F_s(t/r, 1/2), \tag{8.10}$$

obtained by partitioning the index $r\ell + j$. Equation (8.10) holds for every complex s, not only at integer order, and can therefore be differentiated in s.

9 The one-jet kernel for general measures

Atomicity excludes a genuine continuous blind component. Let λ denote Lebesgue measure of mass one on $(0,1)$. Direct integration gives

$$\int_0^1 F_s(t,a)\,\mathrm{d}a = \frac{\zeta(s+1)}{t}, \qquad \Re s > 0. \tag{9.1}$$

Thus a nonzero continuous measure can have both its base tail and every order-derivative tail equal to zero. The next theorem shows that this is the *only* simultaneous blind component.

For a finite complex Borel measure μ on $(0,1)$, its Bernoulli moments $M_n(\mu) = \int B_n(a)\,\mathrm{d}\mu(a)$ always exist. Formal high-tail coefficients are defined by integrating the polynomial coefficients of the single-shift expansion. For an analytic interpretation of both $F_{m,\mu}$ and its first jet, it suffices to assume

$$\int_{(0,1)} \left(1 + |\log a|^2 + |\log(1-a)|^2\right)\,\mathrm{d}|\mu|(a) < \infty. \tag{9.2}$$

Indeed, for $m = 1$ the original series and its first jet grow at most like $1 + |\log a|$ and $1 + |\log a|^2$ near the singular endpoint; these bounds follow by splitting their k-sums at $k \asymp 1/a$. For $m \geq 2$ the corresponding endpoint bounds are no worse. The coefficient theorem below does not require interchanging a measure integral with the initial Mellin contour; such an interchange would require stronger endpoint moment hypotheses.

Theorem 9.1 (Complete one-jet kernel)**.** *Fix $m \geq 1$. On finite complex Borel measures on $(0,1)$, the joint kernel of the two maps*

$$\mu \longmapsto \textit{high tail of } F_{m,\mu}, \qquad \mu \longmapsto \textit{high tail of } \left.\partial_s F_{s,\mu}\right|_{s=m} \tag{9.3}$$

is exactly $\mathbb{C}\lambda$. The same conclusion holds when "zero high tail" is replaced by "eventually terminating high tail." Consequently the two tails determine an arbitrary finite measure modulo a constant multiple of Lebesgue measure, and they determine an atomic measure uniquely.

Proof. The coefficient formula (4.3) and the nonvanishing of B_{2j} show that a zero base tail implies

$$M_{m+2j}(\mu) = 0, \qquad j \geq 1.$$

Independently of any reflection condition, the coefficient in the first jet at t^{m+2j} is a nonzero scalar multiple of $M_{m+2j+1}(\mu)$, because (6.4) is nonzero. Thus a zero first-jet tail also implies $M_{m+2j+1}(\mu) = 0$. Together, these say $M_n(\mu) = 0$ for every $n \geq m+2$. Eventual termination gives the same statement for all sufficiently large n.

Let $\mathcal{E}(z) = \int \mathrm{e}^{az}\,\mathrm{d}\mu(a)$. This is entire, since $0 < a < 1$ and μ is finite. The Bernoulli generating function gives

$$\frac{z\mathcal{E}(z)}{\mathrm{e}^z - 1} = \sum_{n\geq 0} \frac{M_n(\mu)}{n!} z^n = P(z), \qquad |z| < 2\pi,$$

where P is a polynomial. Multiplying through and applying the identity theorem yields

$$\mathcal{E}(z) = \frac{\mathrm{e}^z - 1}{z} P(z) \quad \text{for all } z \in \mathbb{C}, \tag{9.4}$$

with the removable value at $z = 0$ understood. At every nonzero integer n, this gives $\widehat{\mu}(n) = \mathcal{E}(2\pi i n) = 0$. The zeroth coefficient is $\alpha = \mu((0,1))$.

Regarding $(0,1)$ as the circle with its endpoint removed, μ and $\alpha\lambda$ therefore agree on all trigonometric polynomials. Those polynomials are uniformly dense in the continuous functions on the circle. Finite measures that agree on them agree on all continuous functions and hence are identical. Thus $\mu = \alpha\lambda$. Conversely, integrating (2.1) over $a \in (0,1)$ shows $M_n(\lambda) = 0$ for all $n \geq 1$, so all the high tails vanish for λ. Equation (9.1) gives its analytic interpretation. A nonzero finite atomic measure cannot be a multiple of the nonatomic measure λ, proving the last assertion. □

This result is linear injectivity modulo a completely identified one-dimensional component. It does not depend on assuming the base tail is already zero. For a measure in the reflection nullspace, Theorem 7.2 gives the stronger interpretation of the first-jet data as the actual hidden exponential function.

Proposition 9.2 (General-measure reflection classification)**.** *Let $\alpha = \mu((0,1))$. The base tail at order m terminates if and only if*

$$R_*(\mu - \alpha\lambda) = \varepsilon_m(\mu - \alpha\lambda). \tag{9.5}$$

Proof. Integrate the normally convergent Borel kernel (4.7). As in Theorem 8.1, termination forces every residue to vanish. Möbius inversion then makes all positive Fourier coefficients of $\nu = \mu + (-1)^m R_*\mu$ vanish. Its reflection parity supplies the negative coefficients as well. By Fourier uniqueness, $\nu = (1 + (-1)^m)\alpha\lambda$. Rearranging gives (9.5). Conversely, Lebesgue measure contributes no tail and the remaining reflection component cancels the Bernoulli coefficients term by term. □

Theorem 9.1 is therefore sharp in two different senses: no nonzero atomic measure is invisible to the pair of tails, but a Lebesgue component is invisible to every order jet. The explicit integral (9.1) prevents an unjustified extension of atomic injectivity to all finite measures without this quotient.

10 Low-order identities and diagnostic examples

10.1 Order one: a Gamma reflection constant

Using $\zeta'(0,a) = \log\Gamma(a) - \frac{1}{2}\log(2\pi)$ and the Gamma reflection formula, Theorem 5.1 gives

$$\begin{aligned} F_1(t,a) + F_1(t,1-a) = \frac{\pi^2}{3t} - \log(2\sin\pi a) - \frac{t}{2}B_2(a) \\ + 2\sum_{d\geq 1} \frac{\cos(2\pi d a)}{d(\mathrm{e}^{4\pi^2 d/t} - 1)}. \end{aligned} \tag{10.1}$$

This is in the established q-product completion family, not an independent priority claim. Its fixed-parity order derivative has the finite part

$$P_1^{[1]}(t,a) = \frac{2\zeta'(2)}{t} - (\log t + \gamma)\log(2\sin\pi a) - \frac{\zeta''(0,a)+\zeta''(0,1-a)}{2} + tB_2(a)\zeta'(0). \quad (10.2)$$

Here $\zeta'(0) = -\frac{1}{2}\log(2\pi)$. The convergent extra block is

$$\Xi_1(t,a) = 2\sum_{d\geq 1} \frac{\cos(2\pi da)\log(t/(2\pi d))}{d(\mathrm{e}^{4\pi^2 d/t}-1)}. \quad (10.3)$$

Equations (10.2), (10.3), and (7.10) constitute an exact identity for a logarithmically weighted Lambert sum involving Hurwitz-zeta second derivatives.

10.2 A nonzero flat block with a zero base Borel kernel

At $a = 1/2$, (10.1) becomes

$$2F_1(t,1/2) = \frac{\pi^2}{3t} - \log 2 + \frac{t}{24} + 2\sum_{d\geq 1} \frac{(-1)^d}{d(\mathrm{e}^{4\pi^2 d/t}-1)}. \quad (10.4)$$

The normalized base Borel kernel is identically zero. Nevertheless, the last sum begins with $-2Q$, so it is not identically zero. The first order derivative has double poles and its jump is exactly $\pi\mathrm{i}$ times that last sum. At $t = 2$, the hidden remainder is of order 10^{-9}; the delivered verification checks it by subtracting the finite core at 90-digit working precision, not by comparing only power-series coefficients.

10.3 Order two: an antisymmetric identity

The next case has no t^{-1} term:

$$\begin{aligned} F_2(t,a) - F_2(t,1-a) = \zeta(2)(1-2a) - t\big(\zeta'(-1,a) - \zeta'(-1,1-a)\big) + \frac{t^2}{6}B_3(a) \\ - \frac{t}{\pi}\sum_{d\geq 1} \frac{\sin(2\pi da)}{d^2(\mathrm{e}^{4\pi^2 d/t}-1)}. \end{aligned} \quad (10.5)$$

The order-one and order-two formulas exhibit the parity switch: cosines survive in the hidden block for odd m, and sines survive for even m. At m even and $a = 1/2$, the reflected observable is identically zero. This is a zero measure after antisymmetric pairing, not an exception to atomic detection.

10.4 Direct coefficient reconstruction

Suppose an admissible atomic combination has hidden series $\sum b_N Q^N$. The first recovered Fourier coefficients are

$$\begin{aligned} &\widehat{\mu}(1) = b_1, \qquad \widehat{\mu}(2) = 2^m(b_2 - b_1), \qquad \widehat{\mu}(3) = 3^m(b_3 - b_1), \\ &\widehat{\mu}(4) = 4^m(b_4 - b_2), \qquad \widehat{\mu}(6) = 6^m(b_6 - b_3 - b_2 + b_1). \end{aligned}$$

These exact identities separate the divisor convolution from the geometric Fourier problem. For a midpoint grid with $r = 3$, the first two recovered modes vanish and $\widehat{\mu}(3) = -3$; the first action is $12\pi^2$, not $4\pi^2$.

11 Source audit and corrections that matter

11.1 A confirmed cross-reference mismatch

At the pinned commit, the source of [4], near its labels `eq:FR60a` and `eq:FR60b`, describes the equations of Fantini–Rella Conjecture 5 as "(60a)–(60b)" in arXiv:2506.08265v2. The inspected v2 PDF, dated 1 April 2026, displays these as *(3.36a)–(3.36b)*, on its printed page 17 [3]. The proposed correction is to change that external equation reference to (3.36a)–(3.36b), retaining the repository's internal LaTeX labels if desired. This is a version-specific documentary correction, not a refutation of the mathematical identities. The delivery includes the exact target sentence and replacement in a separate note.

11.2 Double poles require a different finite part

A potentially consequential extension error would be to apply ordinary Cauchy principal-value integration directly to (6.5). For a nonzero local coefficient c, the symmetric integral of $c/(\xi-A)^2$ outside an ϵ-neighborhood diverges like $2c/\epsilon$. The Laplace factor does not remove this divergence. The repair is either a Hadamard finite part or the integrated-by-parts simple-pole kernel of Lemma 7.1.

This is a warning for the new order-jet problem. The inspected q-Gamma report [4] uses simple poles and already distinguishes its symmetric kernel from its exponential completion. It is not being accused of making the double-pole error.

11.3 Specialization must not replace analytic differentiation

Theorem 5.1 holds at integer m. Differentiating its right side as though m were a free complex variable is unjustified: there is no specified holomorphic family of the parity sign ε_m, and the finite part changes at resonance. The correct procedure is to differentiate $F_s + \varepsilon_m F_s \circ R$ with the sign fixed, on the initial Mellin contour, and only then specialize. The missing tangent term in (7.11) would be lost by a naive differentiation of the specialized reflection formula. That term is precisely the Borel-summed block of the first jet.

11.4 Other distinctions retained in this report

The zero base Borel kernel in (10.4) coexists with a nonzero flat remainder. This distinction is already explicit in the earlier Lambert and repository literature; it is not a newly discovered error there. Likewise, the linear median used here must not be substituted for an algebra-compatible median in nonlinear products or inverse functions.

The Fourier kernels in this report assume real shifts. For nonreal a, a coefficient such as $\sin(2\pi da)$ can grow exponentially in d, so the displayed Dirichlet expansions generally cease to converge. One cannot claim complex-parameter uniformity by simply inserting complex a into them. Endpoint atoms at $a = 0$ are also outside the stated hypotheses. Finally, the Lebesgue counterexample (9.1) shows why the atomic injectivity theorem requires either atomicity or the explicit quotient in Theorem 9.1.

12 Reproducibility and verification scope

The mathematical arguments above prove infinite identities by contour shifts, functional equations, normal convergence, and finite-dimensional uniqueness. The accompanying program

performs a different task: it checks coefficients, constants, orientations, and representative numerical identities that are especially vulnerable to transcription errors. It is not a proof assistant and it does not establish priority.

12.1 Exact symbolic and arithmetic checks

The delivered run records **732 passing exact assertions** in SymPy 1.14.0. They include Bernoulli reflection cancellation and the activated coefficients for $1 \le m, j \le 8$; rational-kernel residues and double-pole principal parts; the πi normalization through $m = 24$; generic collision jets through derivative order six; Fourier–Möbius calculations in exact cyclotomic fields of moduli 3 through 10; midpoint-grid delay; and the Bernoulli multiplication formula. The free-variable symbolic tests are distinguished in the record from finite arithmetic instances. These computations support, but do not replace, the general proofs.

12.2 High-precision diagnostics

There are **61 passing numerical assertions**, primarily at 90 decimal digits using mpmath 1.3.0. The independent contour tests use 55 digits. The error recorded is

$$\frac{|\mathrm{LHS} - \mathrm{RHS}|}{\max(1, |\mathrm{LHS}|, |\mathrm{RHS}|)}. \tag{12.1}$$

It is a scaled absolute floating-point error, not an interval enclosure. The largest observed errors in the delivered run are summarized below.

Diagnostic	Count	Largest observed error
Exact reflection completion	36	2.46×10^{-91}
Exact first-order-jet completion	10	5.84×10^{-47}
Acceleration refinement	10	5.84×10^{-47}
Independent rotated-contour jump	2	rounded to zero
Independent linear-median kernel	2	5.74×10^{-57}
Nonzero hidden remainder at $a = 1/2$	1	2.55×10^{-91}

The acceptance thresholds were 10^{-75} for reflection and the hidden remainder, 10^{-38} for the full first-jet identities and refinement, and 10^{-42} for the contour tests. A rounded zero in one diagnostic is not a claim of exact numerical equality.

The reflection tests use $m = 1, \dots, 6$, shifts $a = 1/3, 1/2, \sqrt{2}/3$, and $t = 2, 5$. The first-jet tests use $m = 1, \dots, 5$, shifts $1/3, \sqrt{2}/3$, and $t = 3 + m/2$. The original jet is computed from its convergent series with the exact weight $-\log k$, not by differencing nearby orders. The contour tests integrate the rational double-pole kernel on rays of angles ± 0.32 radians at $\lambda = 4, 9$, and compare to (7.3). They check the sign convention independently of the arithmetic summations.

12.3 Why the median evaluation is accelerated

Directly summing (7.10) gives only an N^{-3} tail. For accurate evaluation, subtract finitely many terms of the asymptotic expansion of I from each summand and sum those terms globally using (6.3). For any fixed integer $J \ge 1$, this gives the *exact infinite-sum identity*

$$\begin{aligned} t^m \mathcal{M}\widetilde{J}_m = {} & t^m \sum_{j=1}^{J} d_{m,j}(a) t^{2j} \\ & + \left(\frac{\mathrm{i}t}{2\pi}\right)^{m-1} \sum_{N \ge 1} E_m(N, a) \left(I(A_N/t) - \sum_{j=1}^{J} \frac{(2j)!}{(A_N/t)^{2j+1}} \right). \end{aligned} \tag{12.2}$$

Only the outer N-sum is numerically truncated. The delivered run uses $J = 12$, $N \leq 72$ and compares with $J = 13$, $N \leq 80$. This is finite subtraction, not an assertion that the factorial asymptotic series converges. The numerical truncation and floating-point rounding are not interval-certified; their agreement is a diagnostic.

The general-measure kernel theorem is established by its generating- function proof. The numerical suite does not purport to test all finite measures, unknown support recovery, or complex-shift analytic continuation.

13 Further research questions

13.1 Higher order jets and logarithmic Borel singularities

Corollary 3.2 determines every collision polynomial, but this report gives a complete Borel kernel and exact completion only for the first order derivative. Higher derivatives of $\zeta(s+w)$ introduce polygamma terms, higher derivatives of the functional-equation trigonometric factor, and logarithmic divisor weights. An explicit all-jet completion should be sought with a clear separation of local collision polynomials, lateral discontinuities, and independently identified flat blocks. The first task is to obtain and prove the second-order kernel, rather than infer it by formal specialization.

13.2 Two shifted lattice directions

Replacing the k^{-s} factor by a second shifted lattice sum leads to Mellin products of the form $\Gamma(w)\zeta(w,a)\zeta(s+w,b)$. The two Hurwitz functional equations create a matrix of reflection sectors rather than a single parity. Which sectors terminate, and how many order jets are needed to detect all finite atomic measures on the two-torus? The one-dimensional Lebesgue blind component suggests that continuous product components will have to be identified explicitly.

13.3 Unknown shifts and inverse arithmetic data

Theorem 8.2 gives a finite reconstruction on known nodes. Unknown-node recovery suggests a Prony-type reconstruction from Fourier coefficients obtained by Möbius inversion. Sharp sample counts, treatment of collisions and multiplicities, and certified recovery from approximate Stokes data remain separate problems. Exact identifiability should be settled before numerical stability is claimed.

13.4 Complex shifts and moving endpoints

A kernel holomorphic in a complex neighborhood of a must replace the real Fourier representation; a local meromorphic or integral representation is a plausible route. A second problem is the nonuniform limit in which a approaches 0 or 1 together with t. The fixed-shift Gamma core then changes scale. Neither extension follows from the normal-convergence statements used here.

13.5 Nonlinear inversion after completion

The exact functions N_m and Ξ_m can be transported through a regular inverse or a critical inverse chart. Existing repository work already treats the distinction between formal inversion and inversion of the physically completed function [4]. A useful new application would choose

an actual critical point of a Lambert– polylogarithm combination and compute its first nonzero ramified exponential sectors from the completed forward function. Merely applying a formal inverse theorem to an unspecified flat completion would not settle that problem.

13.6 Formalization and reusable exact certificates

The Bernoulli reflection, divisor convolution, Möbius inversion, Vandermonde uniqueness, and midpoint-grid distribution are finite algebraic targets for formalization. They should be separated from the analytic Mellin shifts and finite-part integrals. A later analytic formalization would need parameter differentiation under normal convergence, a contour-shift theorem with stated growth bounds, and a precise double-pole finite-part convention. The supplied Python checks are not substitutes for those proof-assistant declarations.

Conclusion

Reflection can remove an entire perturbative tail without removing the analytic function's exponentially small remainder. For shifted Lambert–polylogarithm sums, the first order derivative restores exactly the complementary Fourier information, now at double Borel poles. Mellin inversion identifies the derivative's missing logarithmic dual block and proves the factor-πi recovery law. Finite atomic reflection rigidity turns that analytic statement into a finite identification theorem, while the general-measure result isolates Lebesgue measure as the sole simultaneous blind component.

A A compact dependency and status ledger

Result	Main dependencies	Evidence status
All-order collision formula	Mellin differentiation and Laurent coefficients	Proof; finite jet checks
Simple-pole Borel kernel	Bernoulli Fourier expansion and a geometric series	Proof; exact normalization checks
Exact reflected duality	Hurwitz/Riemann functional equations and Mellin inversion	Proof; 36 numerical instances
Double-pole activation	Trivial-zero derivatives and contour orientation	Proof; independent contour tests
Exact first-jet completion	Mellin integration by parts and the I kernel	Proof; 10 numerical instances
Atomic reflection and action bound	Möbius inversion and Vandermonde uniqueness	Proof; exact cyclotomic tests
General one-jet kernel	Bernoulli generating function and Fourier uniqueness	Proof; not numerically certified

No theorem in this ledger is asserted to be Lean-formalized. The literature comparison, source cross-reference check, symbolic tests, numerical diagnostics, and PDF build are distinct evidence scopes. The source report's broader claims about other conjectures are not required by any proof in this article.

B Rebuilding the package

The top-level `Makefile` builds this article with three ordinary pdfLaTeX passes and runs the exact and numerical checks separately. The verification script accepts `-mode exact`, `-mode numerical`, or `-mode all`, and writes only the explicit output path. Its exact dependencies are listed in `requirements.txt`. A typical replay is

```
python verification/verify.py --mode exact \
  --output verification/exact_results.json
python verification/verify.py --mode numerical \
  --output verification/numerical_results.json
make pdf
```

The delivered records were produced with Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0. Replays should be made on a copy of the package if the original recorded outputs are to remain unchanged. The report was prepared with ChatGPT assistance; no independent referee review is implied by the internal proof-review notes.

References

[1] NIST Digital Library of Mathematical Functions. Sections 24.8 (Bernoulli Fourier series), 25.4 (Riemann-zeta functional equation), 25.11 (Hurwitz zeta), and 25.12 (polylogarithms). https://dlmf.nist.gov/24.8, https://dlmf.nist.gov/25.4, https://dlmf.nist.gov/25.11, https://dlmf.nist.gov/25.12. Consulted for the classical identities used in this report.

[2] D. Dorigoni and A. Kleinschmidt, *Resurgent expansion of Lambert series and iterated Eisenstein integrals*, 2020. https://arxiv.org/abs/2001.11035. In particular, the scalar odd-integer degeneration and surviving nonperturbative sectors are discussed in Section 2.2.

[3] V. Fantini and C. Rella, *Modular resurgence, q-Pochhammer symbols, and quantum operators from mirror curves*, arXiv:2506.08265v2, 1 April 2026. https://arxiv.org/abs/2506.08265v2. The inspected Conjecture 5 appears on printed page 17, Eqs. (3.36a)–(3.36b).

[4] ProveIt research report, *Critical Stokes Reversion and q-Gamma Cores*, source dated 4 October 2026, pinned to commit `7bd1ce58abe522b83cd3368bbafa1be97837b75d`. Repository path: `Analysis/Transseries/docs/series-and-transseries/Mellin_Borel_Completion_q_Gamma_Critical_Reversion/critical_stokes_q_reversion.tex`. The inspected ranges include the title/scope, simple-pole kernel, Mellin completion proof, and rational-weight criterion. Repository: https://github.com/VladimirReshetnikov/ProveIt.

[5] V. Reshetnikov and ProveIt contributors, *ProveIt repository documentation*. Inspected paths: `Analysis/Polylogarithms/README.md`, `Analysis/Transseries/README.md`, and `docs/incoming/README.md`, together with the incoming directory listing. Live indexes were read on 12 October 2026; the detailed mathematical comparison is pinned separately in [4]. https://github.com/VladimirReshetnikov/ProveIt.